Shared coefficients and shared baselines in multistate models

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1 Background

A multi-state hazards (MSH) model will model the transitions between multiple states. One example is analysis of the NAFLD data, whose transition diagram is shown in Figure 1; living patients will have 0–3 of the following metabolic comorbidities: diabetes, hypertension, and hyperlipidemia. In this figure there are 10 transitions (arrows). For any given pair of transitions a:b and c:d there are 6 different choices to organize the coefficients and baseline hazards, shown in Table 1. This document explores both the technical and statistical aspects of those choices.

2 Data

We will use two running examples in this document. The first is data on non-alcoholic liver disease (NAFLD) which is represented by the state space shown in Figure 1. Living subjects can have one or more of three metabolic comorbidities of diabetes, hypertension and hyperlipidemia. The study focuses on the impact of a NAFLD diagnosis on the rate at which subjects progress through the three states.

	Baseline hazard		
	Separate	Proportional	Identical
Separate coefficients	1	2	3
Shared coefficients	4	5	6

Table 1: Choices for any pair of transitions in a multistate model.

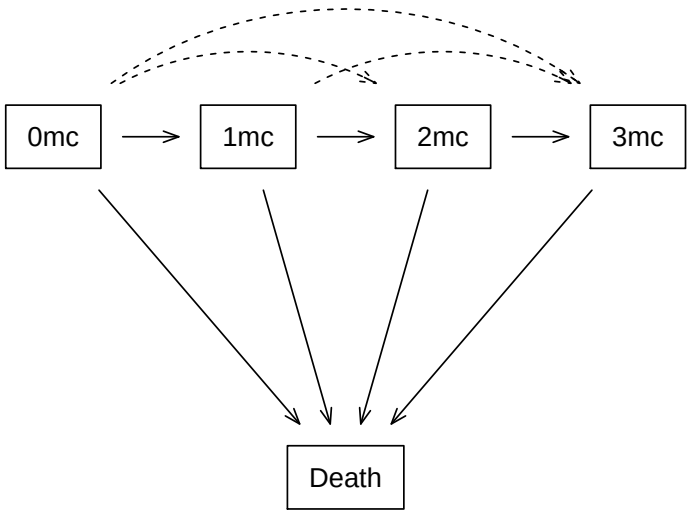


Figure 1: State space for the NAFLD data.

Below is the transition table for the data set. Biologically, a transition directly from 0 to 3 of the comorbidities can not occur; we observe 4 such transitions because our observation of any subject is intermittent. These "jump" transitions, the dotted lines in Figure 1, are a motivation for shared coefficient models.

Note: Results for a MSH fit are in column major order, as is usual for R matrices. The output will have the 0mc:1mc transition, then 0mc:2mc, ..., then 1mc:2mc, ..., 3mc:death.

```
> check1 <- survcheck(Surv(age1, age2, event) ~ nafld + male, data=ndata,
                      id=id, istate=cstate)
> check1$transitions

       to
from     1mc  2mc  3mc death (censored)
  0mc   1829   70    4   263       5705
  1mc      0 1843   28   243       4567
  2mc      0    0 1048   417       3687
  3mc      0    0    0   441       2220
  death    0    0    0     0          0
```

The second example uses the primary biliary cholangitis (PBC) data. The states in this example are created by categorization of the continuous bilirubin variable and the state space is shown in Figure 2. PBC is an inflammatory disease, thought to be of auto-immune origin, characterized by a slow but inexorable loss of the small bile ducts in the liver, as scar replaces functional tissue. Bilirubin and other markers of liver compromise rise slowly over time, then more quickly once excess capacity has been exhausted. Figure 2 shows a multi-state model that includes bilirubin progression. Many of the reverse transitions (dotted lines in the figure) will be subjects whose true bilirubin value is currently near one of the cut points. There are very few observed jumps from normal bilirubin to > 4 or vice-versa (3 and 1, respectively).

```
> pcheck$transitions

        to
from     normal 1-4  >4 death (censored)
  normal      0  90   3    10         77
  1-4        63   0 107    22         60
  >4          1  31   0   108         35
  death       0   0   0     0          0
```

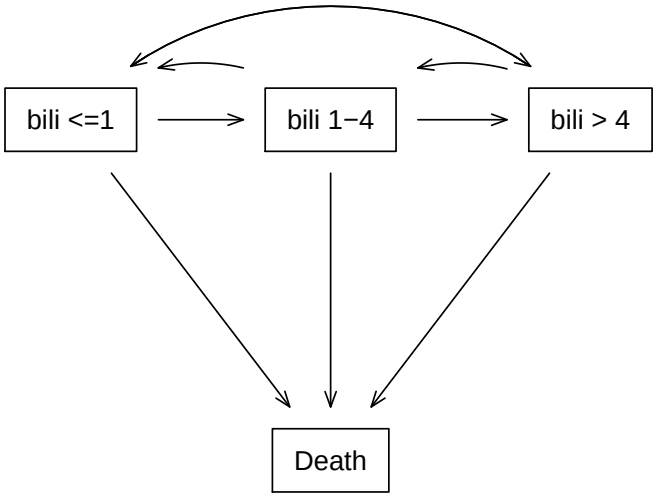


Figure 2: State space for the PBC example.

3 Separate hazards

3.1 Separate coefficients

Separate coefficients and separate hazards for each transition is the default for a multi-state hazards (MSH) model using `coxph`. The code below uses the NAFLD data.

```
> nfit1 <- coxph(Surv(age1, age2, event) ~ nafld + male, data = ndata,
                 id = id, istate = cstate)
> round(coef(nfit1, matrix=TRUE), 2)

       1:2  1:3  2:3   1:4  2:4  3:4  1:5  2:5  3:5  4:5
nafld 0.93 0.31 0.53  1.51 0.08 0.48 0.63 0.53 0.55 0.06
male  0.16 0.54 0.25 12.80 0.02 0.15 0.43 0.47 0.36 0.16
attr(,"states")
[1] "0mc"   "1mc"   "2mc"   "3mc"   "death"

> ## internal mapping matrices
> nfit1$cmap

      1:2 1:3 2:3 1:4 2:4 3:4 1:5 2:5 3:5 4:5
nafld   1   3   5   7   9  11  13  15  17  19
male    2   4   6   8  10  12  14  16  18  20

> nfit1$smap

           1:2 1:3 2:3 1:4 2:4 3:4 1:5 2:5 3:5 4:5
(Baseline)   1   2   3   4   5   6   7   8   9  10

> ## number of current states (cstate) in data
> table(ndata$cstate)

 0mc  1mc  2mc  3mc
7921 6764 5249 2749
```

Two internal matrices `cmap` and `smap` direct the setup of the computation. The first matrix shows the mapping of coefficients to the transitions and the second matrix shows the strata. In this case, there are separate nafld and male coefficients for each transition, and a separate stratum for each. These are used by the internal `stacker` routine to create

expanded data used for the partial likelihood solution. The expanded data has an expanded `X`, `Y` and `transition` variables which contain first all the data rows for transition 1 (0mc:1mc, 7921 rows), then for transition 2 (0mc:2mc, 7921 rows, etc. up to transition 10 (3mc:death, 2749 rows). The transition variable has repeated values of 1, 2, ... 10. The first block of the expanded `Y` has a simple 0/1 status with 1 for a transition from 0mc to 1mc at this point. The expanded `X` matrix is block diagonal for this fit, as shown in equation 1. Here $X^{(1)}$ is all the rows of the data that are at risk for the first transition (7921 rows, 2 columns), and etc. to $X^{(10)}$ for the tenth transition. The final matrix has 20 columns, one for each coefficient, and 65223 rows. The 20 coefficients for the 10 transitions are now fit all at once using a single call to the standard Cox model fitting routine `agreg.fit`, with the transition as a strata.

$$\begin{pmatrix} X^{(1)} & 0 & \dots & 0 \\ 0 & X^{(2)} & \dots & 0 \\ 0 & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & X^{(10)} \end{pmatrix} \tag{1}$$

3.2 Shared coefficients

You will have noticed that `nfit1` gave a warning message the coefficient 8 appears to be infinite, which is the estimated male effect for the 1:4 transition. The estimated risk of exp(12.8) = 362,000 fold for males clearly sticks out. There are only 4 0mc to 3mc transitions in the data, and all of them are males, so indeed the MLE solution is $\hat{\beta} = \infty$; there simply are not enough events in this stratum to estimate 2 coefficients. The other two jumps of Omc:2mc and 1mc:3mc are also quite small.

Biologically, we know that acquisition of 2 new comorbidities on the same exact day does not actually happen. An alternative model is to have coefficients for an increase from 0mc to "one or more" and from 1mc to "two or more". This is shown below.

```
> nfit2 <- coxph(list(Surv(age1, age2, event) ~ nafld + male,
                      1:2 + 1:3 + 1:4 ~ nafld + male / common,
                      2:3 + 2:4 ~ nafld + male/common),
                 data=ndata, id= id, istate= cstate)
> nfit2$cmap

      1:2 1:3 2:3 1:4 2:4 3:4 1:5 2:5 3:5 4:5
nafld   1   1   3   1   3   5   7   9  11  13
male    2   2   4   2   4   6   8  10  12  14
```

```
> round(coef(nfit2, matrix=TRUE), 2)

       1:2  1:3  2:3  1:4  2:4  3:4  1:5  2:5  3:5  4:5
nafld 0.91 0.91 0.52 0.91 0.52 0.48 0.63 0.53 0.55 0.06
male  0.18 0.18 0.25 0.18 0.25 0.15 0.43 0.47 0.36 0.16
attr(,"states")
[1] "0mc"   "1mc"   "2mc"   "3mc"   "death"
```

The final model has 14 coefficients instead of 20, and the overall risk of exp(.18) = 1.2 for males for the 0:mc to 1+ transition is much more sensible. The expanded X matrix in this model will have the same number of rows, but 14 rather than 20 columns. It is no longer block diagonal, column 1 for instance now has non-zero entries in 3 different strata blocks.

The coefficient vector found in the fit object has 14 unique entries and the estimated variance/covariance matrix is 14 by 14, these are what are returned by the `coef` and `vcov` functions. These choices make the MSH model results better fit in to the standard Cox model framework.

3.3 Predicted values

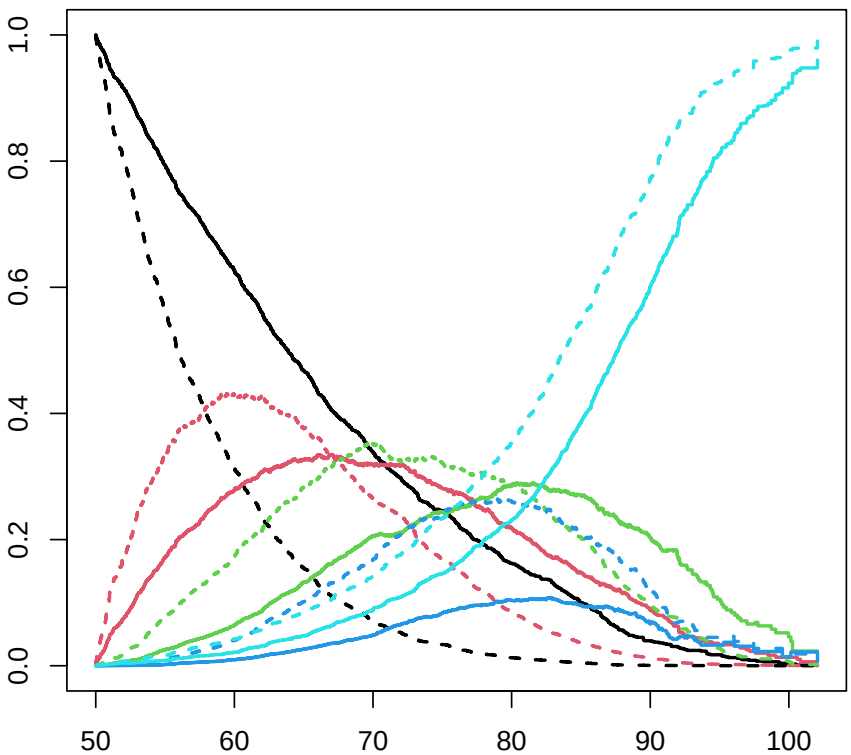
For predicted values from a MSH model, we will more often want to show separate predictions for each transition. For linear predictors, this is facilitated by the `matrix` argument to `coef.coxphms` and `vcov.coxphms`, the MSH methods for the coef and vcov methods. For example

```
> eta2 <- model.matrix(nfit2) %*% coef(nfit2, matrix=TRUE)
> dim(eta2)

[1] 22683    10

> nrow(ndata)

[1] 22683

> v2 <- vcov(nfit2, matrix=TRUE)
> dim(v2)

[1] 14 14
```

The `model.matrix` function for the fit returns the standard X matrix, which is the correct object for this task, not the expanded one which was needed (temporarily) for computing the fit. Then `eta2` is a matrix with one row for each observation in the data set and one column for each transition. This matrix contains more values than we need, formally. Observation 1 for instance is a row for current state 0mc, for instance, so one could argue that the 2mc:death entry of row 1 is not "relevant" to that row of the data.

For predicted probability in state curves, however, we do want all the predicted hazards for a given hypothetical subject, since all the hazards are needed to compute absolute risks.

```
> # predicted curves for a 50 year old female with and without NAFLD, who
> #  starts with none of the comorbidities
> dummy <- data.frame(nafld= 0:1, male=0)
> pstate2 <- survfit(nfit2, newdata=dummy, start.time=50, p0=c(1,0,0,0,0))
> plot(pstate2, col=rep(1:5, each=2), lty=1:2, lwd=2) #, cumprob=TRUE)
> # I don't like this plot, but I'm not sure the alternative below is better
> # plot(pstate2, cumprob=1:4, col=rep(1:4, each=2), lty=1:2, lwd=2)
```



(Yet to do: show examples with predict())

4 Shared proportional hazards

An alternative to separate hazards is shared proportional hazards. We will separate this into three cases: common endpoint will be pairs of transitions such as A:C and B:C to lead to a common state, disjoint a pair like A:C and B:D, and common origin A:B and A:C.

4.1 Common endpoint

This has been by far the most common example in our work. As an example, we might assume that the 4 transitions from comorbidity to death, in the NAFLD data, have the same shape but a different scale. That is

$$\lambda_{1d}(t,x) = \lambda_{0d}(t)\exp(\gamma_1)\exp(X\beta_{.8}) \tag{2}$$
$$\lambda_{2d}(t,x) = \lambda_{0d}(t)\exp(\gamma_2)\exp(X\beta_{.9}) \tag{3}$$
$$\lambda_{3d}(t,x) = \lambda_{0d}(t)\exp(\gamma_3)\exp(X\beta_{.10}) \tag{4}$$
$$\tag{5}$$

where λ_{kd} is the hazard for k comorbidities to death, and we have selected the appropriate column of the coefficient matrix β. For both `nfit1` and `nfit2` the matrix form has has 10 columns 1 per transition, with the last 4 being the transitions to death.

Here is sample code to fit this model, retaining the share coefficients for mc:mc transition of `nfit2`. Behind the scenes, the code bundles the γ and β coefficients together, in order to jointly estimate both of them. We take advantage of the use of "1" as the intercept in the formula modeling language of R along with the fact that the baseline hazard can be thought of as the "intercept" term of a MSH model, for a compact notation.

```
> nfit3 <- coxph(list(Surv(age1, age2, event) ~ nafld + male,
                      1:2 + 1:3 + 1:4 ~ nafld + male / common,
                      2:3 + 2:4 ~ nafld + male/common,
                      0:5  ~ cstate + 1 / common),
                 data=ndata, id= id, istate= cstate)
> print(nfit3, digits=2)

Call:
coxph(formula = list(Surv(age1, age2, event) ~ nafld + male,
    1:2 + 1:3 + 1:4 ~ nafld + male/common, 2:3 + 2:4 ~ nafld +
        male/common, 0:5 ~ cstate + 1/common), data = ndata,
    id = id, istate = cstate)


1:2, 1:3, 1:4  coef exp(coef) se(coef) robust se    z      p
        nafld 0.915     2.496    0.063     0.067 13.6 <2e-16
        male  0.179     1.196    0.046     0.048  3.7  2e-04
```

2:3, 2:4	coef	exp(coef)	se(coef)	robust se	z	p
nafld	0.521	1.683	0.052	0.056	9.3	<2e-16
male	0.246	1.278	0.047	0.050	4.9	9e-07

3:4	coef	exp(coef)	se(coef)	robust se	z	p
nafld	0.485	1.624	0.065	0.068	7.1	9e-13
male	0.149	1.161	0.063	0.065	2.3	0.02

1:5	coef	exp(coef)	se(coef)	robust se	z	p
nafld	0.53	1.70	0.22	0.22	2.4	0.01
male	0.41	1.51	0.12	0.12	3.4	6e-04
cstate1mc	-0.30	0.74	0.13	0.13	-2.4	0.02
cstate2mc	-0.18	0.84	0.12	0.12	-1.5	0.13
cstate3mc	0.62	1.86	0.12	0.12	5.1	4e-07

2:5	coef	exp(coef)	se(coef)	robust se	z	p
nafld	0.51	1.66	0.15	0.15	3.4	6e-04
male	0.47	1.59	0.13	0.13	3.7	2e-04
cstate1mc	-0.30	0.74	0.13	0.13	-2.4	0.02
cstate2mc	-0.18	0.84	0.12	0.12	-1.5	0.13
cstate3mc	0.62	1.86	0.12	0.12	5.1	4e-07

3:5	coef	exp(coef)	se(coef)	robust se	z	p
nafld	0.571	1.771	0.104	0.107	5.3	9e-08
male	0.364	1.438	0.098	0.099	3.7	2e-04
cstate1mc	-0.300	0.741	0.133	0.127	-2.4	0.02
cstate2mc	-0.175	0.839	0.121	0.117	-1.5	0.13
cstate3mc	0.618	1.856	0.123	0.122	5.1	4e-07

4:5	coef	exp(coef)	se(coef)	robust se	z	p
nafld	0.128	1.136	0.098	0.101	1.3	0.21
male	0.160	1.173	0.096	0.099	1.6	0.11
cstate1mc	-0.300	0.741	0.133	0.127	-2.4	0.02

```
  cstate2mc -0.175     0.839    0.121     0.117 -1.5  0.13
  cstate3mc  0.618     1.856    0.123     0.122  5.1 4e-07

 States:  1= 0mc, 2= 1mc, 3= 2mc, 4= 3mc, 5= death

Likelihood ratio test=566  on 17 df, p=<2e-16
n= 22683, unique id= 17549, number of events= 6186
```

Since the current state of each observation is a factor variable `cstate`, the default coding for factors in R does exactly what we wished. We see that after adjustment for sex and NAFLD status, those with 3 comorbidities have a higher death rate than 0mc, which is not a surprise, but also that 1mc appears to be lower than 0mc. More on that will be taken up further below.

One aspect of the printout which is potentially misleading is that all 3 cstate coefficients appear for all 4 transitions. However, for the 1:5 transition (0mc:death) the three 0/1 dummy variables `cstate1mc`, `cstate2mc` and `cstate3mc` will all be 0. The coefficients appear but will never be used. For the 2:5 transition only `cstate1mc` is non-zero and similarly for 3:5 and 4:5. We can get an equivalent model with somewhat simpler printout by creating a single 0/1 dummy variable which is 0 only for the 0mc state.

```
> ndata$notzero <- 1*(ndata$cstate != "0mc")
> nfit3b <- coxph(list(Surv(age1, age2, event) ~ nafld + male,
                       1:2 + 1:3 + 1:4 ~ nafld + male / common,
                       2:3 + 2:4 ~ nafld + male/common,
                       c(2,3,4):5 ~ notzero,
                       0:5  ~ 1 / common),
                  data=ndata, id= id, istate= cstate)
> print(nfit3b, digits=2)

Call:
coxph(formula = list(Surv(age1, age2, event) ~ nafld + male,
    1:2 + 1:3 + 1:4 ~ nafld + male/common, 2:3 + 2:4 ~ nafld +
        male/common, c(2, 3, 4):5 ~ notzero, 0:5 ~ 1/common),
    data = ndata, id = id, istate = cstate)


1:2, 1:3, 1:4  coef exp(coef) se(coef) robust se    z      p
        nafld 0.915     2.496    0.063     0.067 13.6 <2e-16
```

```
        male  0.179     1.196    0.046     0.048  3.7  2e-04


2:3, 2:4  coef exp(coef) se(coef) robust se   z      p
   nafld 0.521     1.683    0.052     0.056 9.3 <2e-16
   male  0.246     1.278    0.047     0.050 4.9  9e-07


3:4      coef exp(coef) se(coef) robust se   z     p
  nafld 0.485     1.624    0.065     0.068 7.1 9e-13
  male  0.149     1.161    0.063     0.065 2.3  0.02


1:5     coef exp(coef) se(coef) robust se   z     p
  nafld 0.53      1.70     0.22      0.22 2.4  0.01
  male  0.41      1.51     0.12      0.12 3.4 6e-04


2:5        coef exp(coef) se(coef) robust se    z     p
  nafld    0.51      1.66     0.15      0.15  3.4 6e-04
  male     0.47      1.59     0.13      0.13  3.7 2e-04
  notzero -0.30      0.74     0.13      0.13 -2.4  0.02


3:5         coef exp(coef) se(coef) robust se    z     p
  nafld    0.571     1.771    0.104     0.107  5.3 9e-08
  male     0.364     1.438    0.098     0.099  3.7 2e-04
  notzero -0.175     0.839    0.121     0.117 -1.5   0.1


4:5        coef exp(coef) se(coef) robust se   z     p
  nafld   0.128     1.136    0.098     0.101 1.3   0.2
  male    0.160     1.173    0.096     0.099 1.6   0.1
  notzero 0.618     1.856    0.123     0.122 5.1 4e-07

 States:  1= 0mc, 2= 1mc, 3= 2mc, 4= 3mc, 5= death

Likelihood ratio test=566  on 17 df, p=<2e-16
```

```
n= 22683, unique id= 17549, number of events= 6186

> all.equal(nfit3$loglik, nfit3b$loglik)

[1] TRUE
```

Another option which provides an even clearer printout is the `shared` option.

```
> nfit3c <- coxph(list(Surv(age1, age2, event) ~ nafld + male,
                       1:2 + 1:3 + 1:4 ~ nafld + male / common,
                       2:3 + 2:4 ~ nafld + male/common,
                       0:5 ~ nafld + male/ shared),
                 data=ndata, id= id, istate= cstate)
> print(nfit3c, digits=2)

Call:
coxph(formula = list(Surv(age1, age2, event) ~ nafld + male,
    1:2 + 1:3 + 1:4 ~ nafld + male/common, 2:3 + 2:4 ~ nafld +
        male/common, 0:5 ~ nafld + male/shared), data = ndata,
    id = id, istate = cstate)


1:2, 1:3, 1:4  coef exp(coef) se(coef) robust se    z      p
        nafld 0.915     2.496    0.063     0.067 13.6 <2e-16
        male  0.179     1.196    0.046     0.048  3.7  2e-04


2:3, 2:4  coef exp(coef) se(coef) robust se   z      p
   nafld 0.521     1.683    0.052     0.056 9.3 <2e-16
   male  0.246     1.278    0.047     0.050 4.9  9e-07


3:4      coef exp(coef) se(coef) robust se   z     p
  nafld 0.485     1.624    0.065     0.068 7.1 9e-13
  male  0.149     1.161    0.063     0.065 2.3  0.02


1:5     coef exp(coef) se(coef) robust se   z     p
  nafld 0.53      1.70     0.22      0.22 2.4  0.01
```

```
  male  0.41      1.51     0.12      0.12 3.4 6e-04


2:5        coef exp(coef) se(coef) robust se    z     p
  nafld    0.51      1.66     0.15      0.15  3.4 6e-04
  male     0.47      1.59     0.13      0.13  3.7 2e-04
  ph(1:5) -0.30      0.74     0.13      0.13 -2.4  0.02


3:5         coef exp(coef) se(coef) robust se    z     p
  nafld    0.571     1.771    0.104     0.107  5.3 9e-08
  male     0.364     1.438    0.098     0.099  3.7 2e-04
  ph(1:5) -0.175     0.839    0.121     0.117 -1.5   0.1


4:5        coef exp(coef) se(coef) robust se   z     p
  nafld   0.128     1.136    0.098     0.101 1.3   0.2
  male    0.160     1.173    0.096     0.099 1.6   0.1
  ph(1:5) 0.618     1.856    0.123     0.122 5.1 4e-07

 States:  1= 0mc, 2= 1mc, 3= 2mc, 4= 3mc, 5= death

Likelihood ratio test=566  on 17 df, p=<2e-16
n= 22683, unique id= 17549, number of events= 6186

> all.equal(nfit3c$loglik, nfit3$loglik)

[1] TRUE
```

The labels in this result more clearly differentiate the γ and β coefficients.

We stated that the baseline hazard plays the role of an intercept, and it does so in another way that is less often noticed. In a linear model, the intercept term makes the fit invariant to simple changes in a covariate, e.g., replace some covariate x in model with $z = x + 2$, and the predicted $\hat{y}$ for each observation remains unchanged as do all of the non-intercept coefficients and their estimate standard errors. A much more common case is replacement of a 0/1 covariate 1/0, e.g., one fit coded 1=male and another 1=female. Compare the coefficients of the modified model below. Nothing changes *except* the ph(1:5) coefficients, nor do they shift by a constant. This means that we must be cautious about

interpreting the absolute values of these coefficients, and perhaps even more any p-values, just as would be the case for a linear model.

```
> ndata$male23 <- ndata$male + 2  # code male as 2/3
> nfit3d <- coxph(list(Surv(age1, age2, event) ~ nafld + male23,
                       1:2 + 1:3 + 1:4 ~ nafld + male23 / common,
                       2:3 + 2:4 ~ nafld + male23/common,
                       0:5 ~ nafld + male23/ shared),
                 data=ndata, id= id, istate= cstate)
> round(coef(nfit3c, matrix=TRUE), 2)

         1:2  1:3  2:3  1:4  2:4  3:4  1:5   2:5   3:5  4:5
nafld   0.91 0.91 0.52 0.91 0.52 0.48 0.53  0.51  0.57 0.13
male    0.18 0.18 0.25 0.18 0.25 0.15 0.41  0.47  0.36 0.16
ph(1:5) 0.00 0.00 0.00 0.00 0.00 0.00 0.00 -0.30 -0.18 0.62
attr(,"states")
[1] "0mc"   "1mc"   "2mc"   "3mc"   "death"

> round(coef(nfit3d, matrix=TRUE), 2)

         1:2  1:3  2:3  1:4  2:4  3:4  1:5   2:5   3:5  4:5
nafld   0.91 0.91 0.52 0.91 0.52 0.48 0.53  0.51  0.57 0.13
male23  0.18 0.18 0.25 0.18 0.25 0.15 0.41  0.47  0.36 0.16
ph(1:5) 0.00 0.00 0.00 0.00 0.00 0.00 0.00 -0.41 -0.08 1.12
attr(,"states")
[1] "0mc"   "1mc"   "2mc"   "3mc"   "death"
```

However, just as in a linear model, this shift does not affect absolute predictions. We point out the following about these fits

1. A predicted curve needs to have a starting time, a starting state, and the covariate values of the hypothetical subject.

2. Covariates associated with the γ coefficients are not a part of newdata. This is because those coefficients belong to the *state*, not to the covariates of the hypothetical subject.

```
> dummy$male23 <- dummy$male + 2
> pstate3c <- survfit(nfit3c, newdata=dummy, start.time=50, p0=c(1,0,0,0,0))
> pstate3d <- survfit(nfit3d, newdata=dummy, start.time=50, p0=c(1,0,0,0,0))
> all.equal(pstate3c$pstate, pstate3d$pstate)
```

```
[1] TRUE
```

The code above recognizes γ coefficient by their association with the special variable name `ph(1:5)`. (I have little doubt that someone, somewhere will eventually create a strange variable enough like this to fool my code, and then post it as a bug.) For probability in state curves based on model `nfit3b` a bit more is needed: the variable `notzero` looks, to the survfit code, no different than any other; the routine needs the user's help to identify that this variable is associated with γ rather than β terms. At this time, the method is to add `notzero` to the dummy data set with a value of NA. From a user interface perspective simply leaving `notused` out of the dummy data set would perhaps be cleaner, but the `model.matrix` routine from base R, which I depend on, is unhappy if a variable found in the model formula is missing from the data. I don't yet have a way to work around that.

To do: example using nfit3b, once I have the NA marker debugged.

To do: add the PBC example.

Summary

One aspect of this that we have not mentioned is that with a set of transitions A:D + B:D + C:D to a common final state, the risk sets also make sense. Since no subject can be two places at once (in two states at the same time), the risk set at time t = (in state A at t) + (in state B at t) + (in state C at t) will have no duplicate subjects. The Cox partial likelihood term for an event at t will have a proper denominator, i.e., a weighted sum over all those at risk.

4.2 Shared hazard, disjoint pairs

The second subcase is a shared baseline hazard for a transition pair like A:B and C:D. In general, it may be hard to think of situations where we would be willing to assume that these two transitions, to two disparate endpoints B and D, would have hazards of the same shape. We have encountered one example, which has motivated the code. It can be seen as an extension of the PBC example.

To do: the example for this is the MCSA data, where I have the CU:dementia transition, then break up CU into substates based on CMC score, a time-dependent covariate, and also break up dementia by CMC score. Almost all the new CU:dementia transitions will be of the form CU+CMC1 to dementia+CMC1 dementia based on CMC score.

One of the important aspects of this example is that the collection of starting states are disjoint, i.e., no given starting state appears more than once (is this true?). Thus the risk sets are proper.

4.3 Shared hazards, common starting state

This is the case where a pair of transitions A:B and A:C are considered to have a common hazard shape. At this point:

- We have not encountered any examples where such an assumption makes biological sense. Without a concrete example, it has been impossible to be completely sure what the software *should* do in this case.
- The current code will create risk sets with multiple copies of the same subject. We are not at all sure that this makes sense.

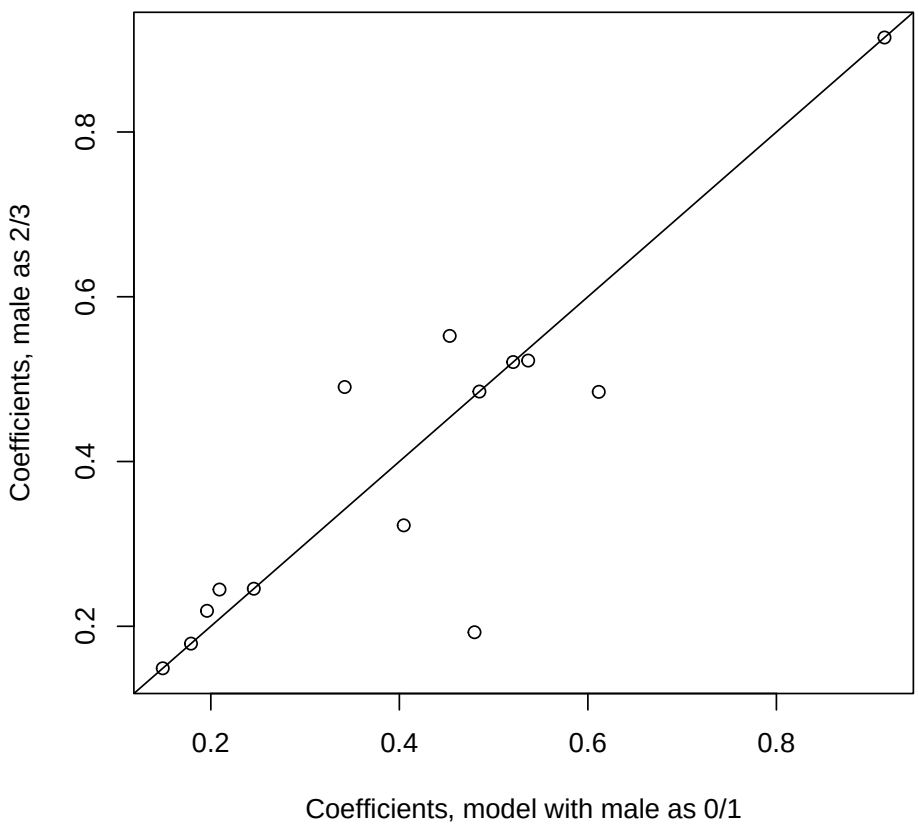
If any reader has such a use case we would be delighted to hear of it.

5 Identical hazards

Identical hazards are a problem child. The problem is that such models can be at the mercy of how covariates are coded. As a simple case, consider the NAFLD data, and assume that the transitions to death have an identical hazard rather than a proportional one. The impact of changing the coding of male sex from 0/1 to 2/3 is profound, with a different log-likelihood and multiple coefficient changes.

```
> nfit4a <- coxph(list(Surv(age1, age2, event) ~ nafld + male,
                       1:2 + 1:3 + 1:4 ~ nafld + male/common,
                       2:3 + 2:4 ~ nafld + male/common,
                       0:5 ~ 1/common),
                  data = ndata, id = id, istate = cstate)
> nfit4b <- coxph(list(Surv(age1, age2, event) ~ nafld + male23,
                       1:2 + 1:3 + 1:4 ~ nafld + male23/common,
                       2:3 + 2:4 ~ nafld + male23/common,
                       0:5 ~ 1/common),
                  data = ndata, id = id, istate = cstate)
> temp <- rbind(male01= nfit4a$log, male23= nfit4b$log)
> colnames(temp) <- c("Initial loglik", "Final loglik")
> temp

       Initial loglik Final loglik
male01      -40534.77    -40284.86
male23      -40534.77    -40260.73
```

```
> plot(coef(nfit4a), coef(nfit4b), xlab="Coefficients, model with male as 0/1",
      ylab= "Coefficients, male as 2/3")
> abline(0,1)
```



Many discussions mention a "common hazard function" as a modeling idea, but few have followed through on the consequences of assuming that $\gamma = 1$ uniformly. If we also force common coefficients for all the covariates then the problem goes away, but that becomes, in essence, a single transition model.